

A REMARK ON THE CONNECTION BETWEEN CHERN-SIMONS-HIGGS THEORY AND TOPOLOGICALLY MASSIVE ELECTRODYNAMICS

FIDEL A. SCHAPOSNIK*

LPTHE, Université Paris 6-7, Tour 24, 5^ét, 2 Place Jussieu, 75251 Paris Cedex 05, France

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We show that the transmutation of the 3-dimensional abelian Chern-Simons-Higgs theory into topologically massive electrodynamics only holds in the topologically trivial sector.

Recently, Deser and Yang¹ discussed the transmutation of a 3-dimensional abelian Chern-Simons (CS) theory with spontaneous symmetry breaking into topologically massive, parity violating electrodynamics.² The connection was established within the $\phi = \phi_0$ vacuum sector (ϕ is the charged scalar and ϕ_0 is the minimum of the Higgs potential). As stressed in Ref. 1, this result may be relevant to the current ideas on high- T_c superconductivity models involving the CS-term. Since topologically non-trivial configurations (vortices) acting as anyons play a central role in these models,³ it is interesting to study whether the connection discovered by Deser and Yang¹ is still valid in topologically non-trivial sectors.

We give, in the present letter, a negative answer to this last question by showing that regular charged vortex configurations, which are known to exist in the Chern-Simons-Higgs model,^{4,5} disappear when the model transmutes into topologically massive electrodynamics. Indeed, as considered in Ref. 1, the action for a charged scalar ϕ coupled to an abelian gauge field A_μ in 3-dimensional space-time with action S given by

$$S = \int d^3x \left(\frac{1}{2} |D_\mu \phi|^2 - \frac{\kappa}{4} \epsilon_{\mu\nu\lambda} A^\mu F^{\nu\lambda} - V[\phi] \right). \quad (1)$$

Note that no Maxwell term is included in S , but solely a CS-term. The potential $V[\phi]$ has a symmetry breaking minimum at $|\phi| = \phi_0$; $D_\mu = \partial_\mu - ieA_\mu$. Deser and Yang¹ have noted that at the (topologically trivial) vacuum $\phi = \phi_0$ the effective action becomes

$$S = \int d^3x (-m^{-1} \epsilon_{\mu\nu\lambda} A^\mu F^{\nu\lambda} + A_\mu A^\mu) \quad (2)$$

with $m = e^2 \phi_0^2 / \kappa$. Now, the “self-dual” first order system described by Eq. (2) has been shown⁶ to be an alternative formulation of the (second order) topologically

* Laboratoire Associé au CNRS UA280.

massive gauge theory in which the Higgs scalar has been traded for an odd-parity helicity-one excitation. In this way a connection between the Chern-Simons-Higgs model and topologically massive electrodynamics was established in Ref. 1. One should note that the authors in Refs. 1 and 6 realized that this equivalence was established in the topologically trivial sector.

Now, it is known that system (1) admits vortex solutions.^{4,5} These are topologically non-trivial static configurations for which the asymptotic behavior of the fields is

$$\lim_{r \rightarrow \infty} \phi = \phi_0 e^{in\varphi}, \quad (3a)$$

$$\lim_{r \rightarrow \infty} A_\mu = \frac{n}{e} \partial_\mu \varphi, \quad (3b)$$

where (r, φ) are polar variables. The topological charge n , $n = 1, 2, \dots$, (related to the quantized magnetic flux of the vortex) is given by

$$\frac{e}{2\pi} \oint dx_\mu A^\mu = n. \quad (4)$$

Due to the presence of the CS-term, the vortex configuration also carries an electric charge Q :

$$Q = -\kappa \left(\frac{ne}{2\pi} \right). \quad (5)$$

When $V[\phi]$ is taken as a sixth-order symmetry breaking potential, vortex solutions can be more easily analyzed since first-order self-dual equations can be shown to exist.^{4,5} Their solution, giving a bound for the Chern-Simons-Higgs system energy, has been studied numerically⁵ and corresponds to regular vortex configurations with magnetic and electric fields concentrated in a ring surrounding the zero of the Higgs field.

What happens if one extends the analysis of Ref. 1 to the topologically non-trivial vacua associated with conditions (3)? In this case, one should consider an effective action for A_μ about the vacuum:

$$\phi^{(n)} = \phi_0 e^{in\varphi}, \quad (6a)$$

$$A_\mu^{(n)} = \frac{n}{e} \partial_\mu \varphi, \quad (6b)$$

associated with the n -topological sector. About it, one still has the effective action, Eq. (2), for A_μ . However, the equations of motion arising from Eq. (2),

$$A_\mu = \frac{m}{2} \varepsilon_{\mu\nu\lambda} \partial^\nu A^\lambda, \quad (7)$$

do not admit regular solutions. Indeed, one can get from Eq. (7) a decoupled equation for A_0 :

$$\Delta A_0 - 4m^2 A_0 = 0, \quad (8)$$

whose vortex-like solution, $A_0 = K_0(2mr)$, is *singular* at the origin. Concerning A_i , $i = 1, 2$, the vortex-like solution to Eq. (7) takes the form

$$A_i = \varepsilon_{ij} \frac{x^j}{r^2} (2mK_1(2mr) - n), \quad (9)$$

which is also singular at the origin. Thus, regular vortex solutions, which are expected to play an important role in anyonic models,³ no longer exist after transmutation of the Chern-Simons-Higgs theory into topologically massive electrodynamics. We then conclude that except for the topologically trivial ($n = 0$) sector, actions (1) and (2) describe different physics.

Let us end by noting that an extension of this result to the non-abelian case is possible: starting from the $SU(N)$ version of action (1) (with N Higgs fields so as to ensure complete symmetry breaking⁷) one can end with the non-abelian version of Eq. (2) by considering different vacuum sectors. Again, topologically non-trivial vortices with Z_N charge are known to exist for the Chern-Simons-Higgs system.⁸ As in the abelian case, they become singular for the topologically massive gauge theory.

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